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Supplemental Information

**Linking the Computational Structure
of Variance Adaptation to Biophysical Mechanisms**

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Supplemental Experimental Procedures.

Model Parameterization

To parameterize the model for fitting, the filter F_{LNK} was defined as a sum of orthonormal basis functions

$$F_{LNK}(t) = \sum_{j=1}^{15} \alpha_j f_j(t + \delta), \quad (12)$$

where α_j are the weighting coefficients and δ is a single delay parameter for each cell that ranged between 0 and 2 ms. The basis functions were decaying sinusoids of increasing frequency

$$f_j(t) = \begin{cases} \sin\left(\pi j \left(\frac{2t}{\tau} - \left(\frac{t}{\tau}\right)^2\right)\right) & 0 \leq t \leq \tau \\ 0 & t > \tau \end{cases}, \quad (13)$$

with τ equal to 1 s and were orthogonalized according to (Keat et al., 2001).

The nonlinearity was a sigmoid function parameterized as:

$$N_{LNK}(x) = a^{\left(\text{erf}(\kappa x + b_1) + 1\right)} k_a^{-1} + b_2 \quad (14)$$

where $\text{erf}(x)$ is the error function defined for cumulative Gaussian distributions, a , b_1 and b_2 are parameters of the model, and κ was a constant equivalent to the overall variance of the filtered stimulus $g(t)$. The exponent of a varied between zero and twenty, and the activation rate constant k_a in the denominator scaled the nonlinearity so that its maximum was one. Thus, the output of the nonlinearity, $u(t)$, scaled the activation rate in the kinetics block between zero and k_a .

For On-Off ganglion cells, two parallel LNK pathways were used with the outputs of each computed as explained above. The outputs were then scaled with independent weights w_{ON} and w_{OFF} and summed to give the final output of the model. The total number of parameters for the model excluding the rate constants set to 0 was 26 for a single pathway and 51 for two parallel pathways.

For numerical integration, a time step of one ms was used for numerical simulation to compute the output of the model. To solve the differential equation for the kinetics block, a discrete-time approximation was used where

$$e^{Q\Delta t} \approx \mathbf{I} + \mathbf{Q}\Delta t \quad (15)$$

for matrix $\mathbf{Q}$ and a small Δt .

Derivation of Instantaneous Gain

We define the instantaneous gain of the kinetics block as the incremental change in the active state, $A(t)$, produced by an incremental change in the input $u(t)$,

$$\text{gain} = \frac{\Delta A}{\Delta u} \quad (16)$$

To compute the gain, we consider the difference between ΔA produced by $u(t)$ alone, and produced by $u(t) + \Delta u$,

$$\frac{\Delta A_{u(t)}}{\Delta t} = k_a u(t) R(t) - k_{fi} A(t) \quad (17)$$

$$\frac{\Delta A_{u(t)+\Delta u}}{\Delta t} = k_a (u(t) + \Delta u) R(t) - k_{fi} A(t).$$

Taking the difference yields the gain

$$\frac{\Delta A_{\Delta u}}{\Delta u} = k_a R(t) \Delta t, \quad (18)$$

indicating that because k_a is constant, the gain is proportional to the resting state occupancy $R(t)$.

Derivation of Adaptive Changes in Temporal Filtering

Given that contrast adaptation depends largely on the mean input to the kinetics block, we examined the effects of an incremental input Δu that is applied on a background of different mean inputs $u_\infty(\sigma)$, which represent a steady input that occurs at a given contrast, σ . Because adaptation changes the kinetics quickly, we consider the effects of a three state system $\{R, A, I\}$ that produces only fast adaptation. Furthermore, because we find empirically that adaptive changes in the time to peak do not depend on the recovery rate constant k_{fr} from the inactive to resting states (Figure 8D), we analyzed a three state system where $k_{fr} = 0$,



We consider the incremental effects $R_\Delta(t)$, $A_\Delta(t)$ and $I_\Delta(t)$ on the state occupancies $R(t)$, $A(t)$, and $I(t)$ produced by a small input Δ . The impulse response from this input is the combined effect of two initial conditions, one with $R \cdot \Delta$ added to the active state, and one with $R \cdot \Delta$ subtracted from the resting state. If we define the magnitude of $R \cdot \Delta$ to be one, then given the initial condition of

$$\{R_{\Delta+}, A_{\Delta+}, I_{\Delta+}\}_{t=0} = \{0, 1, 0\}, \quad (20)$$

the incremental effect on the resting state remains at zero, and the effect on the active state follows the time course

$$A_{\Delta+}(t) = e^{-k_{fi}t}. \quad (21)$$

Given the second initial condition of

$$\{R_{\Delta-}, A_{\Delta-}, I_{\Delta-}\}_{t=0} = \{-1, 0, 0\}, \quad (22)$$

the resting and active states decay as follows,

$$\begin{aligned} R_{\Delta-}(t) &= -e^{-k_a u_\infty(\sigma)t} \\ A_{\Delta-}(t) &= c_2 (e^{-k_a u_\infty(\sigma)t} - e^{-k_{fi}t}), \end{aligned} \quad (23)$$

where $u_\infty(\sigma)$ is the mean input to the kinetics block generated by a constant contrast, σ . The time constant for the incremental response of $R_{\Delta-}(t)$ and $A_{\Delta-}(t)$ are thus a function of contrast. For the active state response, c_2 is a constant. Adding eqs. 21 and 23 for $A_{\Delta+}(t)$ and $A_{\Delta-}(t)$ results in

$$A_\Delta(t) = A_{\Delta+}(t) + A_{\Delta-}(t) = c_2 e^{-k_a u_\infty(\sigma)t} + (1 - c_2) e^{-k_{fi}t}. \quad (24)$$

Thus, the combined effects of the incremental change is a weighted sum of two exponentials, one with a time constant $k_a u_\infty(\sigma)$ that depends on the contrast, and one with time constant k_{fi} that is independent of contrast. The constant c_2 controls whether the kinetics depend on the contrast or not. To solve for c_2 , we combine the equations for $A_{\Delta-}(t)$ and $R_{\Delta-}(t)$ (eq. 23) into

$$\frac{dA}{dt} = k_a R(t) - k_{fi} A(t), \quad (25)$$

and use the fact that $u_\infty(\sigma)$ is a constant to give the result

$$c_2 = \frac{u_\infty(\sigma)}{u_\infty(\sigma) - \frac{k_{fi}}{k_a}}. \quad (26)$$

Thus, when the magnitude of k_{fi}/k_a is small, c_2 approaches one, and weights the variable exponential in eq. 24 more heavily, causing the kinetics of $A_{\Delta}(t)$ to depend on contrast.

Initial Guess and Optimization of the LNK Model

Optimization was done in MATLAB using the function *fmincon*. Since the explicit expression for the error was not available, approximate gradient and Hessian values were used in either an active set or an interior-point algorithm to minimize the given error measure. The only constraints used were boundary constraints for the parameters, which were for the rate constants to be positive, and the slow recovery rate constant k_{sr} to be slower than 1 s. Because there is no explicit formulation for the error surface, the optimum values obtained are assumed to be non-unique and local. To address this issue, we used multiple initial points to converge to different optima and then chose the best solution. Further analysis of the error surface in the neighborhood of the optimal solution is described in the section below.

Initial values for the parameters for the linear filter, F_{LNK} and nonlinearity, N_{LNK} were chosen using the LN model for the high contrast period. Rate constants were chosen randomly between [0,100]. For ON-OFF cells, a principal component analysis was used to obtain the first two stimulus directions for which the cell was most sensitive, (Geffen et al., 2007) and these directions were taken as initial filter estimates. In some cases, we used parameters from previously fit cells as an initial condition. This procedure did not affect the result, but improved the speed of convergence.

To avoid any over-fitting of parameters to the finite dataset, a k-fold cross validation method was used. The data was divided into 5 equal length sets consisting of 300 ms length segments interleaved every 1.5 s. The model was optimized using four of these sets and tested on the fifth. The procedure was then repeated leaving out each of the five sets, and stated correlation coefficient values are the average of the results.

Error Metric

We desired an error metric that would cause the model to fit all segments of the data. However, at different contrasts, the membrane potential shows substantial variation in its amplitude. Thus, using the mean squared error as an error metric would bias the model towards fitting the high contrast segment of the response. In addition, certain segments of the data contain a larger fraction of power in lower frequencies caused by slow shifts in the baseline membrane potential. Thus it was also necessary that the error metric cause the model to capture both slow shifts in

potential and higher frequency fluctuations. In order to address this problem we divided the data into both time and frequency bins, and normalized the error by the standard deviation in each bin. We computed the error measure by first dividing the membrane potential recording $r(t)$ into 10 s time bins, and then filtered each bin in the frequency domain with discrete filters to yield the response, $r_{ij}(t, \omega)$, binned over time interval i and frequency bandwidth j . In practice, only two frequency bins were needed, separated by a cutoff frequency of 4 Hz that divided the response into two bins with approximately equal power. We then computed the error across time and frequency bins as,

$$E = \sum_i \sum_j \frac{\|r_{ij}(t, \omega) - r'_{ij}(t, \omega)\|}{\sigma_{ij}} \quad (27)$$

where σ_{ij} is the standard deviation of the response and $r'_{ij}(t, \omega)$ is the model output for time bin i and frequency bin j . This measure normalized the error in different time bins and frequency components so that no one segment dominated the fitting procedure.

Sensitivity to Parameters

As mentioned above, since the error surface was known to be non-convex, and the obtained optimum solutions local, we tested the sensitivity to parameters in the vicinity of the optimum solutions. Furthermore, we observed whether other parameters could compensate for the increase in error observed after perturbing a single parameter from its optimum value.

To measure the sensitivity of the model to individual parameters, we perturbed five of the parameters of the nonlinearity and kinetics block one at a time from their optimum values and held the parameter constant at the perturbed value (Supplemental Figure 3B). Without allowing any other change in the remaining parameters, we calculated the percentage deviation from the optimum error value. To measure how much the other parameters could compensate for this deviation, the rest of the parameters were reoptimized, and another error value was computed from this new set of parameter values.

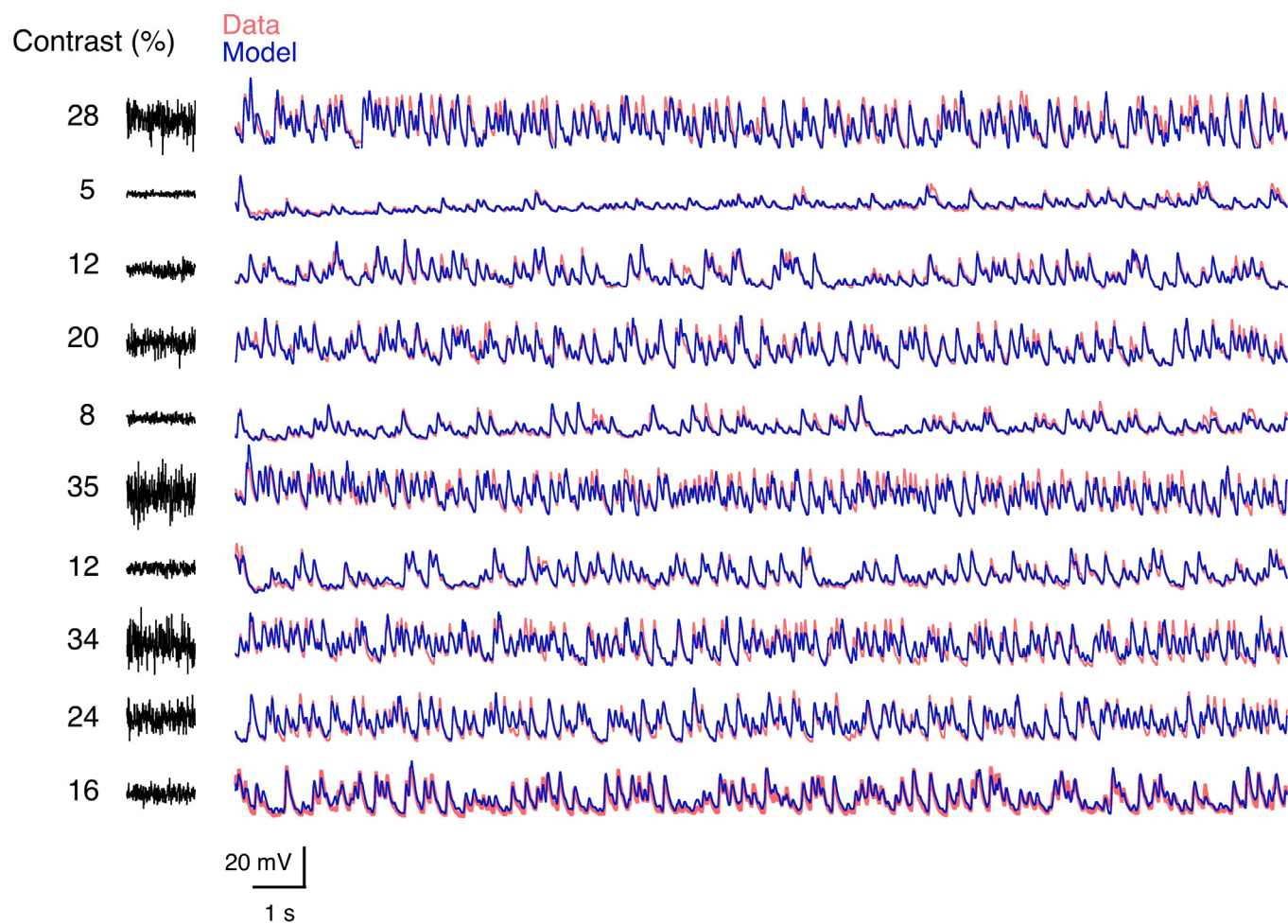
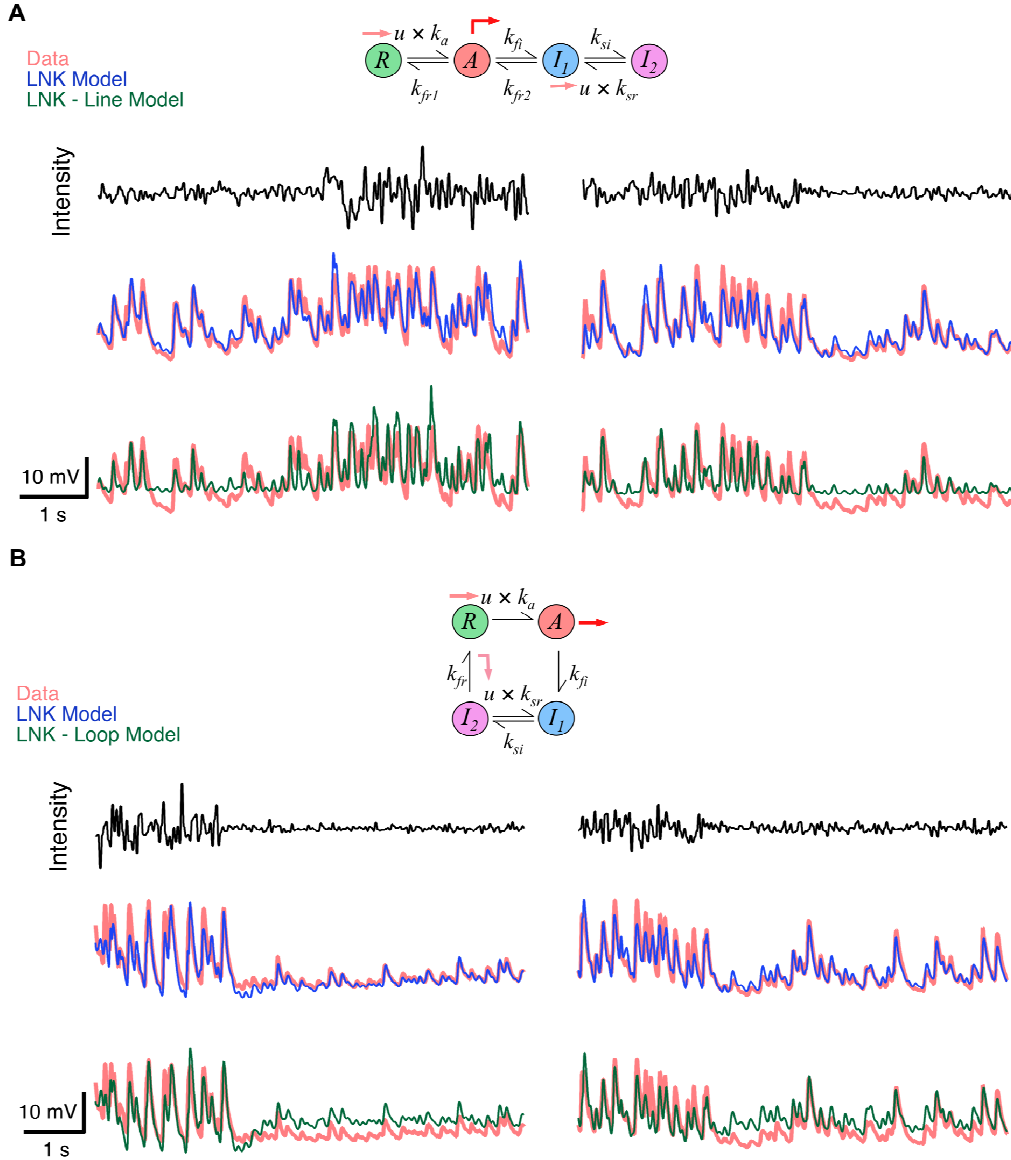


Figure S1. LNK Model of an ON-OFF Ganglion Cell, Related to Figure 2

Comparison of LNK model and membrane potential of an On-Off ganglion cell for 200 s of a continuous recording. Each line is a 20 s segment at a single contrast. Left to right, contrast value, example segment of stimulus intensity during that segment and the membrane potential recording compared to LNK model. Action potentials were digitally removed from the recording.



Supplemental Figure S2. Discarded Topologies for the Kinetics Block, Related to Figure 4

Four-state kinetics block topologies that were discarded because of either lack of slow adaptation or because fast and slow dynamics were coupled, unlike the experimental recording.

(A) Top. Line topology, in which the return pathway from inactivated to resting states is through the active state. Middle. Comparison of a recording with the LNK model shown in the main text. Bottom. Comparison of the same recording with the LNK line-topology model.

(B) Top. Loop topology, in which recovery from the activated state traverses a path with slow rate constants. Middle. Comparison of a recording with the LNK model shown in the main text. Bottom. Comparison of the same recording with the LNK loop-topology model.

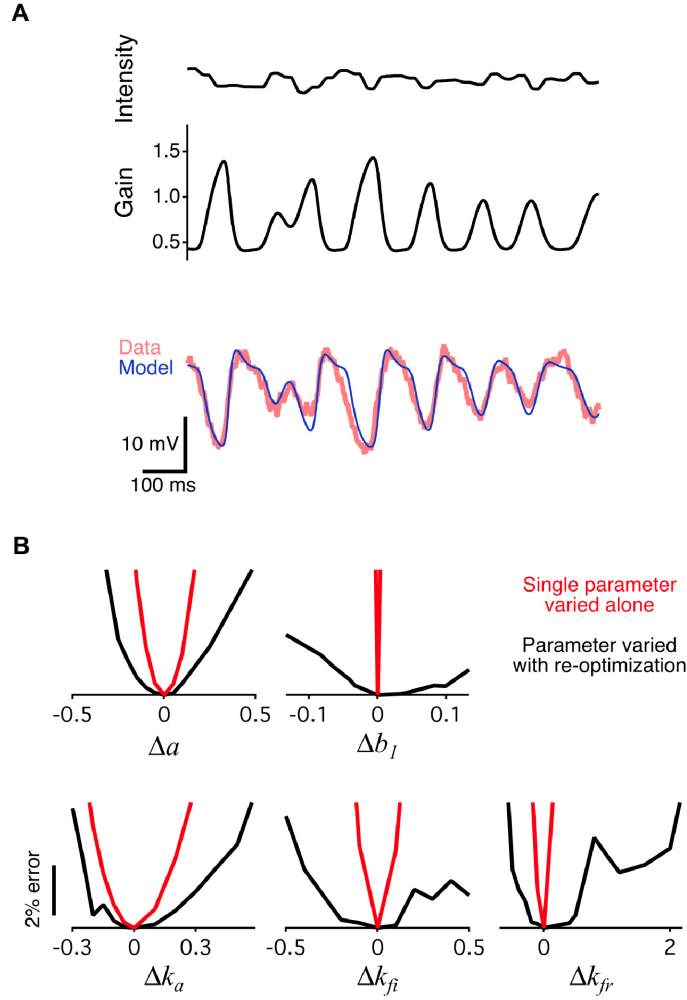


Figure S3. Bipolar Cell Dynamics and Error of Kinetic Parameters, Related to Figure 5

(A) Top. Stimulus intensity at a fixed contrast of 24 %. Middle. For a bipolar cell, the instantaneous gain as represented by the resting state $R(t)$ of the kinetics block. Bottom.

Comparison of the LNK model and membrane potential response.

(B) The error between the LNK model and the recording was computed when single parameters were varied from their optimal value, indicated by zero. The error is the normalized error measure used to fit the model (see Supplemental Experimental Procedures). For each plot, the error value was computed by varying the parameter alone. In addition, the error was computed by holding the parameter fixed at the indicated value and then reoptimizing the model to allow other parameters to compensate for the perturbation. Shown for a single cell are the base of the nonlinearity exponent, a , nonlinearity offset, b_l , maximum activation rate, k_a , fast inactivation rate, k_{fi} and fast recovery rate, k_{fr} . All parameters are normalized by the optimal value of the parameter, and are shown as a fractional change, except for b_l , which was normalized by the full range that the nonlinearity input $g(t)$ spanned by the stimulus. Over 12 cells, to deviate from the optimal error by 2%, the fractional change needed was 0.12 ± 0.02 for a , the exponent of the nonlinearity, 0.13 ± 0.01 for b_l , the input offset of the nonlinearity, where normalization is by the full range of the nonlinearity input, 0.27 ± 0.04 for k_a , the activation rate, 0.23 ± 0.04 for k_{fi} , the fast inactivation rate and 0.24 ± 0.05 for k_{fr} , the fast recovery rate.

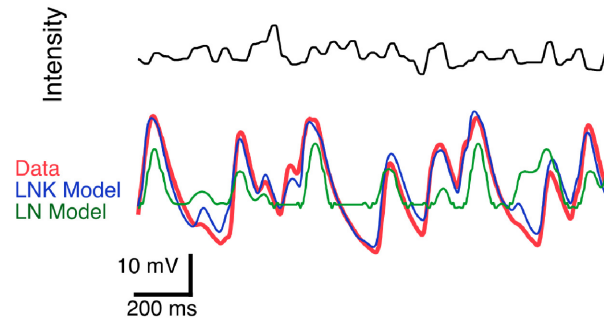
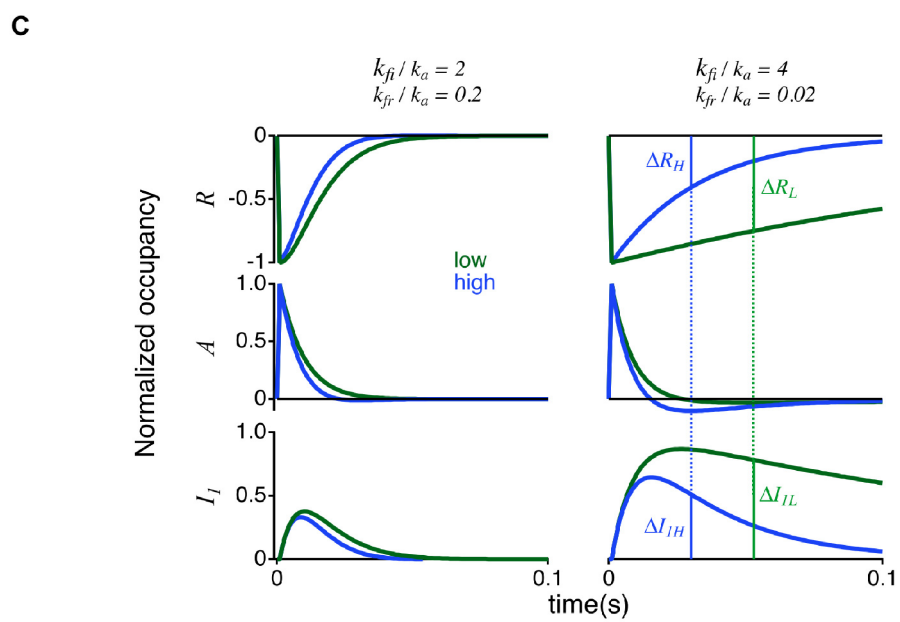
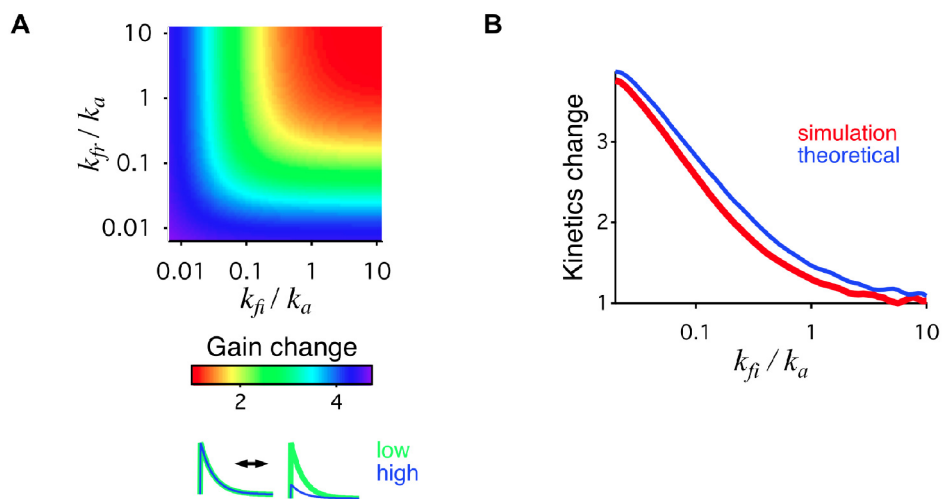


Figure S4. Linear-Nonlinear Model Fails to Capture Ongoing Adaptation and Asymmetric Fast Responses, Related to Figure 7

Top. Stimulus intensity at a fixed contrast of 35 %. Comparison of LN model and LNK model output with the membrane potential recording from an Off-type amacrine cell.



Supplemental Figure S5. Theoretical Estimates for Adaptive Changes, Related to Figure 8

(A) For the kinetics block in Figure 8A, the derived ratio of gain between low and high contrast as a function of two parameters, fast inactivation, k_{fi} and fast recovery, k_{fr} . Gain was computed directly from rate constants using eq. 3. Both parameters are shown normalized by the mean activation rate, k_a .

(B) Theoretical change in kinetics as a function of fast inactivation, k_{fi} normalized by the mean activation rate, k_a . Also shown is the simulation from Figure 8D as a function of k_{fi} . The theoretical change was computed using eqs. 24 and 26. The slight discrepancy is because the fourth state used in the simulation was not accounted for in the theoretical calculation.

(C) The impulse response functions, F_k , for resting, active and inactive (I_1) states were measured by presenting impulses Δu added to a baseline input $u(t)$, representing the average value at low or high contrast. Left. Rate constants were chosen to show a small change in kinetics and temporal differentiation, and show small changes in the level of inactivation as a function of the contrast. Right. Rate constants were chosen to show a larger change in kinetics and temporal differentiation, and show a larger change in the level of inactivation. Dotted lines indicate the trough of the undershoot of the impulse response in the active state. For high contrast, ΔI_{1H} indicates the amount of inactivation at the time of the trough, and ΔR_H indicates the smaller recovered occupancy in the resting state at the time of the trough. Because for a three state model, $\Delta A = -(\Delta I_1 + \Delta R)$, when inactivation ΔI_{1H} is greater than $-\Delta R$, an undershoot in the active state occurs. For low contrast, ΔI_{1L} and ΔR_L indicate corresponding values at the time of the low contrast trough. In this case the change in inactivation is similar to the change in the resting state, and little undershoot occurs in the active state.